

NPU Hardware Evaluation v1.0

A Comparative Study of Edge AI Inference Accelerators

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Abstract

AI inference in production settings is becoming the dominant cost line in enterprise AI. The AI inference market is projected to grow from \$87B in 2024 to \$349B by 2032 (18.9% CAGR) [1]. Neural Processing Units (NPUs), chips built specifically for AI inference, are emerging as a compelling alternative to GPU-only architectures, with 35–70% lower power consumption at comparable throughput [3].

This white paper systematically evaluates ten edge AI inference accelerators across three hardware categories: ASIC NPUs (Hailo-8, Hailo-10H, Axelera Metis, Axelera Europa, EdgeCortex Sakura II), SoC DSPs (SiMa MLSoC, Qualcomm QCS6490, QCS8550), and integrated NPUs (Intel Lunar Lake, AMD XDNA2), benchmarked against an NVIDIA RTX A5000 with TensorRT as a production-grade baseline. Twelve reference models spanning convolutional, mobile and transformer architectures are used as a consistent benchmark suite. Results are analysed for throughput, latency, model compatibility, power efficiency, SDK maturity and product lifecycle.

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Summary: NPUs for Inference

Table 1: Decision criteria for NPU adoption

Criterion	Key signal
TCO & cost per inference	TCO priority jumped from 34% to 41% in a single quarter, overtaking raw performance as the dominant procurement lens. [2]
Power efficiency	NPU-based servers match or exceed GPU throughput while consuming 35–70% less power; no liquid cooling required, cutting electricity costs by 60%+. [3, 5]
Data sovereignty & compliance	On-device inference keeps data off cloud servers, critical under GDPR Art. 25, HIPAA, and DORA (EU banking, live Jan 2025). [6, 7]
Latency for real-time workloads	Autonomous vehicles, industrial automation, and medical imaging require sub-10 ms inference, which cloud round-trips cannot deliver. [8]
Reducing NVIDIA lock-in	With CUDA-based codebases as a strategic liability, firms qualify NPU alternatives to reduce single-vendor dependency. [4, 9]
Software ecosystem maturity	Intel OpenVINO and ONNX Runtime are narrowing the CUDA gap; x86 compatibility remains the enterprise baseline (87% market share, 2025). [10]

Key Findings

Table 2: Platform strengths and weaknesses

Platform	Strengths / Weaknesses
NVIDIA A5000 + TRT	Full model coverage, lowest single-stream latency, mature toolchain. High power (230 W), high cost.
Qualcomm IQ-9075	Best latency among DSPs, full coverage, good SDK. Sequential execution limits throughput.
Hailo-8	Outstanding multi-stream throughput, excellent SDK usability. Limited operator support (4/12 models failed).
Axelera Metis	High peak TOPS, near-linear multi-chip scaling. Same operator coverage limits as Hailo-8.
EdgeCortex Sakura II	Partially different operator support from Hailo/Axelera. BF16 support. Lower multi-stream throughput.
QCS6490 / QCS8550	Full SoC integration, product longevity program. Sequential DSP limits throughput; fragmented SDK.

For full model flexibility with production-grade tooling, the **NVIDIA A5000** remains the reference. For embedded SoC deployments requiring long lifecycle support, **Qualcomm QCS8550 / IQ-9075** represent the strongest option. For workloads supported by the operator sets of ASIC NPUs, the **Hailo-8** and **Axelera Metis** offer the best performance per watt.

Domain-Specific Applications

Healthcare. Protected Health Information (PHI) must not leave the device. NPU-powered real-time diagnostic imaging is projected to grow to a \$2.2B segment by 2027. [11]

Financial services. DORA mandates sovereign audit rights; EU banks are actively repatriating inference workloads on-premise. [7]

Automotive. ADAS and sensor fusion require deterministic sub-10 ms inference; 22M+ vehicles shipped with in-car NPUs in 2023 (22.8% CAGR). [12]

Manufacturing / IIoT. Predictive maintenance and quality control AI; EU Data Act (Sept 2025) expands OT data sovereignty obligations. [13]

Telecom. 72% of base stations integrate programmable NPUs for real-time 5G traffic management. [12]

Enterprise IT. Win 10 EOL refresh wave + Microsoft Copilot+ (40 TOPS minimum) are forcing NPU decisions into standard PC procurement cycles. [14]

1 Project Overview

1.1 Objectives

The primary objective is to systematically collect and consolidate technical know-how on a diverse set of hardware platforms for neural network inference at the edge. The evaluation addresses not only raw performance but also the practical aspects of working with each platform: toolchain quality, optimization workflow difficulty, model support breadth, and long-term hardware viability. The goal is to answer not only which device is fastest, but which device is most practical to adopt, maintain, and build upon.

A central finding that motivates the structure of this report is that peak hardware throughput is rarely the binding constraint in practice. Achieving production ready inference on dedicated accelerators (NPUs in particular) depends critically on how the model is prepared: reducing numerical precision through quantization, introducing weight sparsity, and removing redundant parameters through pruning. Each of these techniques interacts differently with each hardware platform; some accelerators expose dedicated INT4 or BF16

execution units, others require structured sparsity patterns to realize any speedup, and others impose operator constraints that make certain model families altogether incompatible. A fair hardware comparison therefore cannot treat model optimization as a post-hoc concern: it must be evaluated alongside the hardware itself. For this reason, Section 3 provides a grounded treatment of quantization, sparsification, and pruning before the benchmark results are presented.

Concretely, the same model can fail to compile on one NPU, run unoptimized on another, and approach peak hardware efficiency on a third, depending entirely on how it is quantized, sparsified, or pruned beforehand. Framing the optimization techniques explicitly in Section 3 allows the benchmark results that follow in Section 4 to be read not merely as a leaderboard, but as a reflection of each platform’s ability to exploit reduced-precision arithmetic and sparse or compact computation.

1.2 Hardware Platforms

Three categories of inference hardware were evaluated.

ASIC NPUs are dedicated silicon devices designed for neural network inference, implementing fixed, highly optimised dataflows and memory hierarchies. This specialisation enables exceptional performance-per-watt but at the cost of flexibility: supported operations, quantization formats, and network topologies are constrained by the fixed architecture. Evaluated devices: Hailo-8, Hailo-10H, Axelera Metis, Axelera Europa, and EdgeCortex Sakura II.

SoC DSPs integrate a DSP or DSP array within a broader SoC alongside general-purpose CPU cores, memory controllers, and often additional accelerator blocks. DSPs offer greater programmability than pure ASIC solutions, and their integration within a full SoC makes them well-suited for complete embedded systems. Evaluated devices: SiMa MLSoC, Qualcomm QCS6490, and Qualcomm QCS8550.

Integrated NPUs are neural network acceleration blocks embedded directly within general-purpose consumer or workstation processors, sharing the same package and memory subsystem as the host CPU and GPU. Evaluated devices: Intel Lunar Lake (Core Ultra Series 2) and AMD XDNA2 (Ryzen AI 300 series).

As a performance reference baseline, an **NVIDIA RTX A5000** GPU was included, with all inference executed through TensorRT compiled in INT8 mode.

1.3 Evaluation Methodology

Each platform was assessed across the following dimensions:

1. **Installation and demo inference.** End-to-end hardware and software setup, producing a step-by-

step installation guide and, where applicable, a reproducible container environment.

2. **SDK-based network import, optimization, and quantization.** Full workflow from a floating-point model to a deployable, optimized artefact.
3. **Model zoo analysis.** Breadth, documentation quality, and practical utility of vendor-provided model repositories.
4. **Optimization assessment on reference models.** Exercise of the full optimization and quantization pipeline on twelve reference architectures.
5. **Inference application development.** A minimal but complete inference application using the vendor’s native SDK: model loading, raw tensor input (blob in), raw tensor output (blob out), and FPS measurement.
6. **Performance analysis.** Throughput (FPS) relative to rated peak performance (TOPS) to derive an efficiency metric.
7. **Product lifecycle.** Announced/expected product lifetime, SDK maturity and update frequency, supply chain continuity.

1.4 Reference Models

A fixed set of twelve neural network models was selected to cover the major architectural families in modern computer vision (Table 3).

Table 3: Reference benchmark models

Model	Family	Params
ResNet-18	ResNet	11.69M
ResNet-34	ResNet	21.80M
ResNet-50	ResNet	25.56M
ResNet-101	ResNet	44.57M
ResNeXt-101 32×8d	ResNeXt	88.79M
EfficientNet-B0	EfficientNet	5.29M
MobileNetV3-S-100	MobileNet	2.54M
MobileNetV4-Conv-S	MobileNet	3.77M
MNASNet-0.75	MNASNet	2.91M
ConvNeXt-Tiny	ConvNeXt	28.59M
ViT-Tiny p16/224	ViT	5.72M
Tiny-ViT-5M/224	Hybrid ViT	5.39M

All models operate on 224×224 RGB inputs and are drawn from established, publicly available model repositories, primarily sourced from the PyTorch Image Models (**timm**) library. [15]

2 Hardware Platform Specifications

Note on platform coverage. All platforms described in this section are included for completeness and situational awareness. However, benchmarking results (Section 4) are available only for the platforms that have been fully tested to date: Hailo-8, Axelera Metis, EdgeCortex Sakura II, and the Qualcomm DSP family (QCS6490, QCS8550, IQ-9075). The **Hailo-10H**, **Intel Lunar Lake**, and **AMD XDNA2** platforms are currently under evaluation; benchmark data for these devices will be reported in a future revision of this document. The **Axelera Europa** was not yet available for hardware evaluation at the time of writing; benchmarking will be conducted once hardware access is confirmed. The **SiMa MLSoC** was not benchmarked due to SDK integration constraints described in Section 2; evaluation remains planned pending improved software support.

2.1 ASIC NPUs

2.1.1 Hailo-8

The Hailo-8 delivers up to 26 TOPS at a typical power consumption of 2.5 W, making it one of the most power-efficient ASIC NPUs at the time of its release. A defining architectural feature is its fully integrated on-chip memory, which eliminates the need for external DRAM entirely. The chip is offered both as an M.2 module, exposing a PCIe Gen 3.0 interface (2 lanes in B+M key, 4 lanes in M-key), and as a HHHL PCIe card for server and industrial-PC deployments. Supported frameworks include TensorFlow, PyTorch, ONNX and Keras. [17]

2.1.2 Hailo-10H

The Hailo-10H delivers 40 TOPS of INT4 performance (equivalent to 20 TOPS INT8) with exceptional power efficiency. Unlike the Hailo-8, it includes a direct DDR interface for large models (LLMs, VLMs, Stable Diffusion). The module carries 8 GB of LPDDR4 on-module memory and connects via a 4-lane PCIe Gen 3 interface, available in M.2 Key M (2242/2280) form factor as well as a HHHL PCIe card variant, with typical chip power below 3.5 W. [17]

Evaluation status: *The Hailo-10H has not yet been benchmarked. Hardware and SDK setup is in progress; results will be included in a future revision.*

2.1.3 Axelera Metis

The Axelera Metis is built around a proprietary RISC-V-based architecture employing Digital In-Memory Computing (D-IMC), delivering up to 214 TOPS of INT8 performance at 15 TOPS/W. It is offered both as a single-slot HHHL PCIe Gen 3 x4 card and as an M.2

2280 module, available in 4 GB and 16 GB DRAM configurations. The Voyager SDK supports TensorFlow, PyTorch, and ONNX. [19]

2.1.4 EdgeCortex Sakura II

The Sakura II is built on EdgeCortex’s second-generation Dynamic Neural Accelerator (DNA-II) architecture, delivering 60 TOPS of INT8 and 30 TFLOPS of BF16 within a typical power envelope of 8 W. It is available both as a HHHL PCIe Gen 3 (x8/x16) card and as an M.2 2280 module. The memory subsystem uses a dual 64-bit channel LPDDR4x interface with up to 16 GB on-board (peak bandwidth 68 GB/s), complemented by 20 MB of on-chip SRAM. The MERA software suite supports PyTorch, TensorFlow Lite, and ONNX. [20]

2.1.5 Axelera Europa

The Axelera Europa is the second-generation AIPU from Axelera AI, delivering 629 TOPS of INT8 performance within a 45 W TDP. The chip integrates eight second-generation AIPU cores alongside 16 RISC-V vector processing units for on-chip pre- and post-processing, 128 MB of L2 SRAM, and a memory subsystem delivering 200 GB/s external DRAM bandwidth. Supported precision formats include INT4, INT8, and INT16. The Europa is supported by the same Voyager SDK used for the Metis, enabling models trained in PyTorch, TensorFlow, and ONNX to be compiled and deployed across the full Axelera platform without SDK changes. [16]

Evaluation status: *The Axelera Europa is currently in early access; hardware was not yet available during the evaluation period covered by this report. Benchmarking will be conducted once hardware access is confirmed; results will be included in a future revision.*

2.2 SoC DSPs

2.2.1 SiMa MLSoC

The SiMa MLSoC integrates a 50 TOPS machine learning accelerator, a quad-core Arm Cortex-A65 application processor, a Synopsys EV74 computer vision DSP, H.264/H.265 hardware codec, 4 MB on-chip memory, a 32-bit quad-channel LPDDR4 memory controller, and 8 lanes of PCIe Gen 4, all on a 16 nm TSMC process node. The Palette software provides a unified compilation and deployment stack. [21]

Evaluation status: *The SiMa MLSoC has not been benchmarked in this evaluation. At the time of testing, SiMa’s inference SDK exclusively exposed a GStreamer-based pipeline interface, which is incompatible with our tensor-level benchmarking framework (raw blob in / blob out). Evaluation will be revisited once the SDK offers a direct tensor I/O path; results will be included in a future revision.*

Table 4: ASIC NPU platform specifications

Device	Peak TOPS	Power	Memory	Interface	Precision
Hailo-8	26 (INT8)	2.5 W	None (on-chip)	PCIe Gen 3 x2/x4	INT8
Hailo-10H	40 (INT4) / 20 (INT8)	<3.5 W	8 GB LPDDR4	PCIe Gen 3 x4	INT4, INT8, FP16
Axelera Metis	214 (INT8)	8–15 W	4–16 GB DRAM	PCIe Gen 3 x4	INT8
Axelera Europa	629 (INT8)	45 W	128 MB L2; 200 GB/s BW	PCIe (TBD)	INT4, INT8, INT16
Sakura II	60 (INT8) / 30 TFLOPS (BF16)	8 W	16 GB LPDDR4x	PCIe Gen 3 x8/x16	INT8, BF16

2.2.2 Qualcomm QCS6490

The QCS6490 is a 6 nm SoC integrating an octa-core Qualcomm Kryo 670 CPU, an Adreno 643 GPU, and a Qualcomm Hexagon AI Engine (DSP + HVX + Hexagon Tensor accelerator) delivering up to 12 TOPS. It is part of Qualcomm’s Product Longevity Program with supply commitment until July 2036. AI inference is accessed via the AI Engine Direct SDK and SNPE/QNN runtimes. [22]

2.2.3 Qualcomm QCS8550

The QCS8550 is Qualcomm’s flagship IoT and embedded processor, built on 4 nm, featuring an octa-core Kryo CPU (peak 3.36 GHz), an Adreno 740 GPU, and an 8th-generation Hexagon AI Engine delivering 48 TOPS of INT8 performance. Like the QCS6490, it is part of Qualcomm’s Product Longevity Program. [22]

2.2.4 Qualcomm Dragonwing IQ-9075

The IQ-9075 is the entry point of Qualcomm’s Dragonwing IQ9 industrial SoC series, pairing an octa-core Kryo Gen 6 (Cortex-A78C-based) CPU clocked at 2.1–2.36 GHz with an Adreno 663 GPU and dual Hexagon Tensor Processors. AI throughput is offered in two SKU variants: 50 dense TOPS (QCS9075-AC) and 100 dense TOPS (QCS9075-AA), with sparse-computing configurations on some reference boards quoted up to 200 TOPS@INT8. The platform supports up to 36 GB of LPDDR5 (6×16-bit channels @ 3200 MHz) with inline ECC, and exposes two PCIe Gen 4 links (2-lane and 4-lane). Like the QCS6490 and QCS8550, the IQ-9075 is covered by Qualcomm’s Product Longevity Program, with industrial-grade operation from −40 to +115 °C (junction) and a committed supply horizon exceeding 10 years. AI workloads are accessed via the AI Engine Direct SDK and QNN/SNPE runtimes, with framework support for TensorFlow, PyTorch, and ONNX. [23]

2.3 Integrated NPUs

2.3.1 Intel Lunar Lake (Core Ultra Series 2)

The Intel Lunar Lake NPU4 architecture delivers up to 48 TOPS NPU-alone, a threefold increase over the

preceding Meteor Lake NPU3 generation. A key departure is the integration of on-package LPDDR5X memory (16 GB or 32 GB), reducing memory latency and cutting system power by ≈40% versus off-package solutions. Total platform AI performance across CPU, GPU, and NPU reaches 120 TOPS. Intel retained FP16 support, whereas competing solutions from AMD and Qualcomm top out at INT8. The NPU is accessible via the OpenVINO toolkit and the Windows AI platform (DirectML / ONNX Runtime). [24]

Evaluation status: *The Intel Lunar Lake NPU has not yet been benchmarked. Hardware and SDK setup is in progress; results will be included in a future revision.*

2.3.2 AMD XDNA2 (Ryzen AI 300 Series)

The AMD XDNA2 NPU is a spatial dataflow architecture with 32 AI Engine tiles, delivering up to 50 TOPS of INT8 performance, a claimed 5× compute and 2× power efficiency improvement over the first-generation XDNA. A notable addition is Block FP16 support, which performs computations at 8-bit integer throughput with 16-bit floating-point numerical accuracy. Off-chip memory bandwidth reaches up to 120 GB/s (dual-channel LPDDR5); on-chip SRAM achieves 800 GB/s for AI engine tiles. The NPU is accessible via AMD’s Ryzen AI software stack and ONNX Runtime. [14]

Evaluation status: *The AMD XDNA2 NPU has not yet been benchmarked. Hardware and SDK setup is in progress; results will be included in a future revision.*

2.4 Reference Baseline: NVIDIA RTX A5000

The RTX A5000 (NVIDIA Ampere) features 8,192 CUDA cores, 256 third-generation Tensor Cores, and 24 GB of GDDR6 ECC memory on a 384-bit bus (768 GB/s bandwidth). Single-precision performance is 27.8 TFLOPS; Tensor Core performance reaches 222.2 TFLOPS (FP16 with sparsity). TDP is 230 W via PCIe 4.0 x16. All inference in this evaluation is executed through TensorRT compiled in INT8 mode. [25]

Table 5: SoC DSP platform specifications

Device	Peak AI TOPS	Process	CPU	AI Engine	Memory
SiMa MLSoC	50	16 nm	4× Cortex-A65 @ 1.15 GHz	ML Accel. + EV74 DSP	LPDDR4 (ext.)
QCS6490	12 (INT8)	6 nm	8× Kryo 670	Hexagon DSP + HVX + Tensor	LPDDR5 (ext.)
QCS8550	48 (INT8)	4 nm	8× Kryo (peak 3.36 GHz)	8th-gen Hexagon + Tensor	LPDDR5 (ext.)
IQ-9075	50/100 (dense)	—	8× Kryo Gen 6 @ 2.1–2.36 GHz	Dual Hexagon HTP (HVX + HMX)	Up to 36 GB LPDDR5

Table 6: Integrated NPU platform specifications

Device	NPU TOPS	Platform TOPS	Package TDP	Memory	Precision
Intel Lunar Lake	48	120	17–30 W	16–32 GB LPDDR5X (on-pkg)	INT8, FP16
AMD XDNA2	50	—	28–54 W	LPDDR5 (external)	INT4, INT8, BF16, Block FP16

Table 7: NVIDIA RTX A5000 specifications

Specification	Value
CUDA Cores	8,192
Tensor Cores	256 (3rd generation)
Tensor Performance	222.2 TFLOPS (FP16 w/ sparsity)
Memory	24 GB GDDR6 ECC
Memory Bandwidth	768 GB/s
TDP	230 W
Host Interface	PCIe 4.0 x16
Inference Runtime	TensorRT (INT8)

2.5 Quick Reference Summary

Table 8 shows a summary of all the evaluated platforms in a side-by-side comparison. Qualcomm SoC power is system-dependent and not directly comparable to standalone accelerator TDP figures.

3 Model Optimization Techniques

As deep learning models continue to scale in size and complexity, deploying them efficiently on dedicated hardware such as NPUs requires the application of model optimization techniques. Three complementary strategies bridge the gap between model accuracy and deployment efficiency: *quantization*, *sparsification*, and *pruning*.

3.1 Quantization

Quantization reduces the numerical precision used to represent a model’s weights and activations. Standard training operates in FP32; quantization maps these values to lower-precision formats (most commonly INT8, but increasingly INT4, INT3, or even INT2).

This is beneficial for two primary reasons: (1) lower-precision data types require less memory for storing model parameters and activations; (2) integer arithmetic is substantially cheaper than floating-point on most hardware. NPUs in particular are often optimized with dedicated integer execution units delivering dramatically higher throughput for quantized workloads.

3.1.1 Quantization Hyperparameters

Static quantization computes all scale factors and zero points once using a calibration dataset, achieving the best performance in throughput, latency, and memory. **Dynamic quantization** computes these on-the-fly during inference, yielding better accuracy but significantly reducing practical benefits.

Quantization can be **symmetric** (zero point fixed at zero) or **asymmetric** (unconstrained zero point). Asymmetric quantization better exploits the integer range when activation distributions are shifted, as after ReLU. It can also be applied **per-tensor** (single scale factor) or **per-channel** (separate scale per output channel).

3.1.2 Calibration

Calibration determines appropriate min-max ranges for each layer’s outputs, defining scale factors and zero points. Methods include:

Table 8: Side-by-side specifications of all evaluated platforms.

Device	Category	Peak TOPS	Typical Power	Memory	Interface
Hailo-8	ASIC NPU	26 (INT8)	2.5 W	None (on-chip)	PCIe Gen 3 x2/x4
Hailo-10H	ASIC NPU	40 (INT4) / 20 (INT8)	<3.5 W	8 GB LPDDR4	PCIe Gen 3 x4
Axelera Metis	ASIC NPU	214 (INT8)	8–15 W	4–16 GB DRAM	PCIe Gen 3 x4
Axelera Europa	ASIC NPU	629 (INT8)	45 W	128 MB L2; 200 GB/s BW	PCIe (TBD)
Sakura II	ASIC NPU	60 (INT8)	8 W	16 GB LPDDR4x	PCIe Gen 3 x8/x16
SiMa MLSoC	SoC DSP	50	≈5 W	LPDDR4 (external)	PCIe Gen 4 x8
QCS6490	SoC DSP	12 (INT8)	SoC-level	LPDDR5 (external)	Integrated SoC
QCS8550	SoC DSP	48 (INT8)	SoC-level	LPDDR5 (external)	Integrated SoC
IQ-9075	SoC DSP	50 / 100 (dense)	SoC-level	Up to 36 GB LPDDR5	PCIe Gen 4
Intel Lunar Lake	Integrated NPU	48 NPU / 120 platform	17–30 W (pkg)	16–32 GB LPDDR5X	Integrated SoC
AMD XDNA2	Integrated NPU	50 (INT8)	28–54 W (pkg)	LPDDR5 (external)	Integrated SoC
NVIDIA RTX A5000	GPU (reference)	222 TFLOPS (tensor)	230 W	24 GB GDDR6 ECC	PCIe 4.0 x16

- *MinMax*: uses observed minimum and maximum values.
- *MovingAverageMinMax*: smooths min-max estimates via a running average.
- *HistogramMSE*: minimises the MSE between FP32 and quantized outputs.
- *HistogramEntropy*: minimises KL-divergence between distributions.
- *HistogramPercentile*: clips outliers at a chosen percentile.

A poor calibration choice can cause accuracy drops as large as 10%. There is no universally optimal method.

3.1.3 Overflow Handling

Mitigation strategies include saturation (compute at higher bitwidth then saturate-cast), overflow-aware calibration, overflow-aware QAT, and stochastic rounding (shown empirically to reduce quantization bias).

3.1.4 Post-Training Quantization and QAT

Post-training quantization (PTQ) converts an already-trained FP32 model to lower precision using a calibration dataset, without retraining. It can be enhanced with techniques such as AdaRound. **Quantization-aware training (QAT)** simulates quantization effects during training via straight-through estimators, allowing the model to learn representations robust to reduced precision. QAT is typically introduced gradually and followed by a final PTQ

step.

3.2 Sparsification

Sparsification introduces zeros into weight matrices or activation tensors. By itself it offers limited benefits; its value lies primarily as a prerequisite for pruning or, when backed by hardware support, as a route to measurable inference speedups.

3.2.1 Types of Sparsity

Weights can be selected using **magnitude-based** (zeros out smallest-absolute-value weights; simple and efficient) or **gradient-based** criteria (selects by importance w.r.t. the loss; more informed but costlier). Sparsity structure varies from **unstructured** (individual weights, requires hardware support), **structured** (entire filters/channels, compatible with standard dense hardware), to **semi-structured** (regular blocks, a middle ground).

3.2.2 Hardware Support for Sparsity

Block Floating Point (BFP) represents weights in blocks sharing a common exponent, allowing the hardware to skip computation when a full block is zero. **N:M sparsity patterns** (e.g., 2:4 supported in NVIDIA Ampere) require exactly N out of every M consecutive weights to be non-zero, enabling compressed weight storage and doubled throughput through dedicated sparse execution units. [26]

3.2.3 Enforcing Sparsity in Practice

Sparsification is typically implemented by replacing standard layers with sparse-aware equivalents carrying a binary mask applied before each forward pass. The mask is updated after each training epoch, with sparsity introduced gradually until the target level is reached. Structured sparsity requires consistency across layer boundaries.

3.3 Pruning

Pruning permanently removes zeroed-out parameters or structural components. The result is a smaller, more compact architecture that requires fewer parameters and less computation, with benefits independent of hardware sparse execution support.

3.3.1 Weight and Structured Pruning

Weight pruning removes individual parameters based on a saliency criterion and fine-tunes the sparse model. **Structured pruning** eliminates entire neurons, convolutional filters, or attention heads driven to zero by structured sparsification. Because the pruned model retains a regular, dense layout, it is immediately usable with standard hardware without specialized sparse support.

Pruning entire layers is feasible only in residual networks, since a zero-weight layer can be removed as long as the residual connection preserves the signal path.

3.3.2 Practical Pruning Process

The most effective approach is an iterative cycle: train to convergence, prune a small percentage of the lowest-importance weights, retrain to recover accuracy, and repeat. There is no universal recipe; empirical experimentation remains necessary.

3.4 Combined Application and Joint Optimization

In practice the three techniques are rarely applied in isolation. Their interactions often offer synergies: pruning reduces model complexity, stabilizing subsequent quantization; quantization can push small residual weights to exactly zero, further increasing sparsity; a pruned, sparse model presents a simpler calibration landscape.

Quantization-aware pruning uses an importance score incorporating both weight magnitude and quantization error. **Iterative quantization and pruning** alternates between both steps during training. **Joint loss functions** incorporate both a quantization error term and a sparsity penalty directly into the training objective.

From a hardware evaluation perspective, NPUs differ significantly in how well they exploit quantized arith-

metic, handle sparse computation, and benefit from structured versus semi-structured sparsity. The benchmarks presented in the following sections are designed to surface these differences.

4 Results and Analysis

This section presents benchmark results across all evaluated platforms, covering throughput and latency on the twelve reference models (Section 1), multi-stream and multi-chip throughput scaling on the Hailo-8, Axelera Metis, and EdgeCortex Sakura II, quantization strategy effects on the NVIDIA A5000, and a cross-platform discussion of key findings.

4.1 Cross-Platform Throughput

Tables 9 and 10 report latency (ms) and throughput (FPS) for all twelve reference models across ASIC NPUs, SoC DSPs, and the A5000 reference baseline. **FAIL** indicates a model could not be compiled or executed due to unsupported operators or memory constraints.

For the Hailo-8, only single-stream throughput is shown in Table 9; multi-stream pipelined throughput is reported separately in Table 12. No equivalent multi-stream scaling is available on the Qualcomm Hexagon DSP platforms, which execute one inference job at a time.

The most immediate finding is the significant variation in model compatibility across device categories. The NVIDIA A5000 with TensorRT successfully compiled and executed all twelve models. The Qualcomm DSP family achieved near-complete coverage (the sole exception being MobileNetV3 on QCS8550). The ASIC NPUs exhibited substantial compatibility gaps: Hailo-8 failed on four models (ConvNeXt-Tiny, EfficientNet-B0, MobileNetV3-Small, and Tiny-ViT-5M), Axelera Metis showed the same failure set plus ResNeXt-101 and ViT-Tiny, and EdgeCortex Sakura II failed on ConvNeXt-Tiny, EfficientNet-B0, ViT-Tiny, and Tiny-ViT—though it succeeded on MobileNetV3 where both Hailo and Axelera failed. The consistent failure of all three ASIC NPUs on transformer-based models highlights a fundamental limitation of fixed-function silicon, with growing implications as model architectures diversify.

4.2 Axelera Metis Multi-Chip Scaling

The quad-chip Axelera Metis PCIe card (856 TOPS peak) was tested on the subset of models that compiled successfully on the single-chip variant. As shown in Table 11, throughput scales near-linearly across standard convolutional architectures. MNASNet-0.75 scales

Table 9: ASIC NPU throughput vs. NVIDIA RTX A5000 (TensorRT INT8). **FAIL** = model could not be compiled or run.

Model	Hailo-8		Axelera Metis		EdgeCortex S2		A5000 TRT	
	Lat.	FPS	Lat.	FPS	Lat.	FPS	Lat.	FPS
convnext_tiny	FAIL	FAIL	FAIL	FAIL	FAIL	FAIL	0.87 ms	1218
efficientnet_b0	FAIL	FAIL	FAIL	FAIL	FAIL	FAIL	0.57 ms	1898
mnasnet_075	1.87 ms	3081	2.0 ms	1156	1.02 ms	956	0.27 ms	4481
mobilenetv3_small	FAIL	FAIL	FAIL	FAIL	0.83 ms	1162	0.34 ms	3393
mobilenetv4_conv_s	1.28 ms	5948	1.5 ms	1581	0.76 ms	1262	0.24 ms	5299
resnet18	1.20 ms	2509	1.8 ms	1131	1.09 ms	894	0.20 ms	6800
resnet34	4.02 ms	1139	2.5 ms	580	1.56 ms	630	0.32 ms	3738
resnet50	5.02 ms	771	2.9 ms	516	1.94 ms	507	0.40 ms	2844
resnet101	7.17 ms	112	4.6 ms	330	2.79 ms	355	0.69 ms	1554
resnext101_32x8d	15.17 ms	61	FAIL	FAIL	14.32 ms	71	0.86 ms	1236
tiny_vit_5m_224	FAIL	FAIL	FAIL	FAIL	FAIL	FAIL	0.71 ms	1515
vit_tiny_p16_224	9.66 ms	94	FAIL	FAIL	FAIL	FAIL	0.48 ms	2317

Table 10: SoC DSP throughput vs. NVIDIA RTX A5000 (TensorRT INT8). **FAIL** = model could not be compiled or run.

Model	QCS6490		QCS8550		IQ-9075		A5000 TRT	
	Lat.	FPS	Lat.	FPS	Lat.	FPS	Lat.	FPS
convnext_tiny	6.40 ms	150	4.59 ms	450	1.57 ms	645	0.87 ms	1218
efficientnet_b0	2.30 ms	305	3.44 ms	720	0.69 ms	1503	0.57 ms	1898
mnasnet_075	1.30 ms	398	1.82 ms	928	0.33 ms	3377	0.27 ms	4481
mobilenetv3_small	1.20 ms	446	FAIL	FAIL	0.31 ms	3523	0.34 ms	3393
mobilenetv4_conv_s	1.00 ms	444	1.49 ms	951	0.27 ms	4092	0.24 ms	5299
resnet18	2.30 ms	344	1.48 ms	954	0.50 ms	2122	0.20 ms	6800
resnet34	3.20 ms	261	1.84 ms	795	0.76 ms	1365	0.32 ms	3738
resnet50	3.30 ms	238	2.34 ms	690	0.91 ms	1148	0.40 ms	2844
resnet101	5.00 ms	174	3.63 ms	557	1.42 ms	714	0.69 ms	1554
resnext101_32x8d	10.50 ms	87	20.10 ms	270	3.39 ms	296	0.86 ms	1236
tiny_vit_5m_224	6.10 ms	133	16.70 ms	257	3.03 ms	334	0.71 ms	1515
vit_tiny_p16_224	3.90 ms	217	5.81 ms	420	1.52 ms	666	0.48 ms	2317

from 1,156 FPS to 4,622 FPS ($4.00\times$); MobileNetV4-Conv-S achieves the same $4.00\times$ factor. Heavier ResNet backbones range from $3.68\times$ (ResNet-101) to $4.60\times$ (ResNet-34), reflecting differences in workload-partitioning efficiency across model sizes. For workloads within Axelera’s operator support envelope, the multi-core card represents a compelling path to very high throughput in a compact, low-power form factor.

4.3 Hailo-8 Multi-Stream Throughput

The Hailo-8 implements a pipelined dataflow architecture that partitions a neural network across fixed hardware stages. When six inference jobs are submitted concurrently, they overlap in pipeline fashion, yielding a near- $6\times$ increase in throughput at unchanged per-request latency. Table 12 reports single-stream

(FPS@1) and six-stream (FPS@6) figures for all models that compiled successfully. Throughput scales almost exactly $6\times$ for lightweight convolutional models; the reduced multipliers observed for ResNet-101 ($5.31\times$) and ResNeXt-101 ($5.48\times$) are a direct consequence of on-chip memory constraints discussed in Section 4.6.

4.4 EdgeCortex Sakura II Dual-Chip Throughput

EdgeCortex Sakura II was also evaluated in a dual-chip configuration (two chips partitioned together). Table 13 compares single-chip and dual-chip latency and throughput for all models that completed deployment in both configurations. Unlike the Axelera Metis, which scales near-linearly up to $4.60\times$ with four cores, the Sakura II dual-chip configuration delivers a

Table 11: Axelera Metis 1× vs. 4× throughput scaling

Model	Lat.	FPS@1	FPS@4	Scale
mnasnet_075	2.0 ms	1156	4622	4.00×
mobilenetv4_conv_s	1.5 ms	1581	6331	4.00×
resnet18	1.8 ms	1131	4552	4.03×
resnet34	2.5 ms	580	2668	4.60×
resnet50	2.9 ms	516	2211	4.28×
resnet101	4.6 ms	330	1215	3.68×

Table 12: Hailo-8 single-stream vs. six-stream pipelined throughput

Model	Lat.	FPS@1	FPS@6	Scale
mnasnet_075	1.87 ms	3081	18485	6.00×
mobilenetv4_conv_s	1.28 ms	5948	35684	6.00×
resnet18	1.20 ms	2509	15053	6.00×
resnet34	4.02 ms	1139	6838	6.00×
resnet50	5.02 ms	771	4625	6.00×
resnet101	7.17 ms	112	595	5.31×
resnext101_32x8d	15.17 ms	61	334	5.48×
vit_tiny_p16_224	9.66 ms	94	567	6.03×

more modest throughput gain of 1.13–1.33×, suggesting that inter-chip communication or workload-partitioning overhead partially offsets the additional compute capacity.

It should be noted that the dual-chip partition mode evaluated here is not the only deployment option when two Sakura II chips are available. Rather than joining both chips into a single larger partition for one network, each chip can instead be programmed independently to run a *different* network. This enables two fully independent inference pipelines to run in parallel, each with single-chip throughput, effectively doubling the number of concurrent models that can be served. For applications such as multi-task vision systems or sensor-fusion pipelines that require separate networks running simultaneously, this independent partition mode may be more practical than the throughput scaling mode benchmarked above.

4.5 NVIDIA A5000 Quantization Strategy

Tables 14–16 break down the effect of quantization format on latency, throughput, and top-1 accuracy for the A5000 with TensorRT. Table 17 reports the fraction of operations executed per precision for selected models, illustrating where hard versus soft calibration diverges.

The transition from FP32 to FP16 already provides substantial gains for convolutional architectures: ResNet-18 improves from 1,508 to 4,335 FPS (2.9×). Moving to INT8 provides a further step: ResNet-18 reaches 6,800 FPS under INT8 Weak (4.5× over FP32). ModelOpt-assisted INT8 consistently outperforms both

Weak and Hard INT8, delivering 20–60% gains over standard INT8 for convolutional models.

Transformer architectures exhibit markedly different behaviour. For Tiny-ViT-5M and ViT-Tiny, INT8 Hard calibration *increases* latency relative to INT8 Weak, because it forces 94.5% and 96.5% of operations to FP32 respectively (Table 17). Soft calibration retains attention operations in FP16 while achieving INT8 coverage for the remaining layers. ModelOpt recovers throughput and achieves the best result for transformers (2,615 FPS for ViT-Tiny).

Accuracy losses are modest for convolutional architectures (<0.5% top-1 degradation for ResNet/ResNeXt). Notable outliers are EfficientNet-B0 (≈3.9 percentage-point loss under INT8 Weak) and MobileNetV3-Small, which loses 13.8 points under INT8 Hard and a severe 53 points under ModelOpt INT8—a warning that depthwise-heavy architectures with squeeze-and-excitation blocks may require mixed-precision strategies.

4.6 Discussion

4.6.1 Hailo Pipelining vs. DSP Sequential Execution

The Hailo-8 implements a pipelined dataflow architecture in which a neural network is partitioned into stages assigned to dedicated hardware blocks. Multiple inference jobs overlap in pipeline fashion, analogous to CPU instruction pipelining at the inference level. The practical consequence is illustrated directly in Table 12: for MNASNet-0.75, single-stream throughput is 3,081 FPS, while six-stream pipelined throughput

Table 13: EdgeCortix Sakura II 1× vs. 2× chip throughput

Model	1×		2×		Scale
	Lat.	FPS	Lat.	FPS	
mnasnet_075	1.02 ms	956	0.99 ms	1270	1.33×
mobilenetv3_small	0.83 ms	1162	0.70 ms	1459	1.26×
mobilenetv4_conv_s	0.76 ms	1262	0.65 ms	1461	1.16×
resnet18	1.09 ms	894	0.83 ms	1016	1.14×
resnet34	1.56 ms	630	1.08 ms	787	1.25×
resnet50	1.94 ms	507	1.49 ms	653	1.29×
resnet101	2.79 ms	355	2.12 ms	464	1.31×
resnext101_32x8d	14.32 ms	71	12.67 ms	80	1.13×

Table 14: Latency (ms) by quantization format on NVIDIA RTX A5000 with TensorRT

Model	FP32	FP16	INT8 Weak	INT8 Hard	INT8 ModelOpt
convnext_tiny	1.23	0.87	0.87	1.09	0.76
efficientnet_b0	0.78	0.64	0.57	0.57	0.40
mnasnet_075	0.36	0.29	0.27	0.27	0.16
mobilenetv3_small	0.41	0.34	0.34	0.35	0.28
mobilenetv4_conv_s	0.32	0.24	0.24	0.24	0.13
resnet18	0.71	0.28	0.20	0.20	0.12
resnet34	1.32	0.47	0.32	0.31	0.22
resnet50	1.22	0.53	0.40	0.40	0.27
resnet101	2.27	0.95	0.69	0.69	0.51
resnext101_32x8d	2.52	1.20	0.86	0.86	0.67
tiny_vit_5m_224	1.00	0.70	0.71	1.21	0.75
vit_tiny_p16_224	0.76	0.48	0.48	1.09	0.38

reaches 18,485 FPS—a 6× multiplier that closely tracks the number of concurrent pipeline stages.

The Qualcomm Hexagon DSP operates sequentially: the entire DSP fabric is occupied by a single inference job from input to output, and a second job cannot begin until the first has completed. No comparable scaling is achievable regardless of concurrent requests. This distinction has major implications for video analytics pipelines or batch inference services prioritizing aggregate throughput.

4.6.2 Single-Stream Latency

When only single-stream latency is considered, the picture reverses. The A5000 consistently achieves the lowest latency, often 3–10× lower than the ASIC NPUs. For ResNet-18, the A5000 achieves 0.20 ms versus 1.20 ms (Hailo-8) and 1.8 ms (Axelera Metis). The Qualcomm IQ-9075 occupies a useful middle ground, reaching 0.50 ms for ResNet-18 and approaching GPU-class latency for lightweight models. The Hailo-8’s relatively high single-stream latency is a direct consequence of its pipelined architecture: a single request must traverse the full pipeline depth before producing output.

4.6.3 Platform-Level Takeaways

NVIDIA (A5000 + TensorRT). Full model coverage across all quantization configurations, highest single-stream throughput, and a mature toolchain. Real-world multi-stream throughput benefits are more limited than the raw TOPS figure suggests, as batch scheduling overhead and memory bandwidth constraints become bottlenecks at high concurrency.

Qualcomm (QCS6490, QCS8550, IQ-9075). Near-complete model coverage and competitive single-stream latency; the IQ-9075 approaches GPU-class latency for lightweight models. Sequential execution prevents pipelined throughput scaling. The SDK ecosystem is functional but fragmented, with multiple SDK generations (SNPE, QNN, AI Engine Direct) adding integration overhead.

Hailo-8. Outstanding multi-stream throughput within its operator support envelope, with FPS@6 figures competitive with or exceeding the A5000 for several lightweight convolutional models. The compiler and runtime APIs are notably well-designed. The primary limitation is operator coverage: four of twelve benchmark models could not be compiled. A discrepancy was

Table 15: Throughput (FPS) by quantization format on NVIDIA RTX A5000 with TensorRT

Model	FP32	FP16	INT8 Weak	INT8 Hard	INT8 ModelOpt
convnext_tiny	846	1219	1218	959	1298
efficientnet_b0	1356	1692	1898	1918	2450
mnasnet_075	3229	4138	4481	4466	5895
mobilenetv3_small	2734	3450	3393	3318	3503
mobilenetv4_conv_s	3775	5170	5299	5227	7221
resnet18	1508	4335	6800	6739	8236
resnet34	790	2354	3738	3763	4519
resnet50	854	2079	2844	2855	3631
resnet101	450	1108	1554	1550	1929
resnext101_32x8d	405	867	1236	1238	1468
tiny_vit_5m_224	1045	1529	1515	865	1314
vit_tiny_p16_224	1409	2301	2317	964	2615

Table 16: Top-1 accuracy (%) by quantization format. ONNX column shows the floating-point reference.

Model	ONNX ref.	FP32	FP16	INT8 Weak	INT8 Hard	INT8 ModelOpt
convnext_tiny	55.28	55.28	55.31	55.18	55.59	55.27
efficientnet_b0	95.23	95.23	95.28	91.36	91.36	83.78
mnasnet_075	66.21	66.26	66.19	65.96	65.96	67.18
mobilenetv3_small	86.06	86.06	86.03	85.29	72.22	32.81
mobilenetv4_conv_s	86.19	86.24	86.16	86.98	86.98	87.59
resnet18	94.87	94.87	94.87	94.98	94.98	94.72
resnet34	95.94	95.94	96.00	95.87	95.87	95.99
resnet50	97.37	97.37	97.35	97.24	97.24	97.07
resnet101	96.68	96.68	96.71	96.73	96.73	96.48
resnext101_32x8d	88.73	88.73	88.76	88.38	88.35	89.45
tiny_vit_5m_224	94.16	94.14	94.16	90.98	90.85	90.72
vit_tiny_p16_224	80.02	80.02	80.02	80.02	79.97	79.10

observed between our measured throughput for ResNet-34 and ResNet-50 and the figures reported in official Hailo benchmarks [18]. We attribute this to an architectural difference between the PyTorch model weights used in this evaluation (sourced from `torchvision`) and the TensorFlow-derived variants used in Hailo’s published results: the TensorFlow variants appear to reduce channel width more aggressively in early stages, resulting in a lighter compute graph. The exact reason of the difference is not fully confirmed, but readers should keep this in mind when comparing these specific figures against vendor-published numbers.

A further constraint is the Hailo-8’s fixed on-chip SRAM budget. The device carries no external DRAM; all activations and weights must reside entirely on-chip. Larger models such as ResNet-101 and ResNeXt-101 exceed this budget and must be partitioned by the compiler into multiple execution contexts, each of which is loaded and dispatched sequentially. This sequential multi-context execution breaks the assumptions of the pipelined dataflow architecture: rather than a single

continuous pipeline stage, the hardware processes several independent sub-graphs in sequence, stalling between them. The performance penalty is clearly visible in Tables 9 and 12: while lightweight models achieve near- $6\times$ six-stream scaling, ResNet-101 and ResNeXt-101 reach only $5.31\times$ and $5.48\times$ respectively. The same constraint also depresses single-stream throughput for these models—ResNet-101 reports 112 FPS compared to 771 FPS for ResNet-50, a drop disproportionate to the $1.7\times$ difference in parameter count. This on-chip memory ceiling is therefore an important practical consideration when targeting the Hailo-8 with deeper or wider architectures.

Axelera Metis. Highest peak TOPS among single-chip ASIC NPUs evaluated, with near-linear multi-chip throughput scaling (Section 4.2). Operator coverage mirrors Hailo-8 and represents the same fundamental constraint.

EdgeCortex Sakura II. Partially different operator coverage from Hailo and Axelera, succeeding on Mo-

Table 17: Fraction of operations executed per precision format (selected models)

Model + Format	FP32	FP16	INT8
convnext_tiny – FP32	100.0%	0.0%	0.0%
convnext_tiny – FP16	4.7%	95.3%	0.0%
convnext_tiny – INT8 Weak	4.6%	94.6%	0.8%
convnext_tiny – INT8 Hard	72.5%	0.0%	27.5%
efficientnet_b0 – FP32	100.0%	0.0%	0.0%
efficientnet_b0 – FP16	0.5%	99.5%	0.0%
efficientnet_b0 – INT8 Weak	1.7%	0.0%	98.3%
efficientnet_b0 – INT8 Hard	1.7%	0.0%	98.3%
resnet18 – FP32	100.0%	0.0%	0.0%
resnet18 – FP16	4.2%	95.8%	0.0%
resnet18 – INT8 Weak	8.0%	0.0%	92.0%
resnet18 – INT8 Hard	8.0%	0.0%	92.0%
tiny_vit – FP32	100.0%	0.0%	0.0%
tiny_vit – FP16	1.2%	98.8%	0.0%
tiny_vit – INT8 Weak	1.2%	89.0%	9.8%
tiny_vit – INT8 Hard	94.5%	0.0%	5.5%
vit_tiny – FP32	100.0%	0.0%	0.0%
vit_tiny – FP16	2.2%	97.8%	0.0%
vit_tiny – INT8 Weak	2.2%	97.8%	0.0%
vit_tiny – INT8 Hard	96.5%	0.0%	3.5%

bileNetV3 where both ASIC competitors failed. Dual-chip throughput scaling is moderate (1.13–1.33×, Table 13), lower than the near-linear scaling of Axelera Metis, but BF16 support and larger on-board DRAM make it more flexible for mixed-precision and generative AI workloads.

4.6.4 Optimisation Takeaways

INT8 quantization is effectively a solved problem for standard convolutional architectures: PTQ with appropriate calibration recovers near-lossless accuracy while delivering 2–5× throughput improvements over FP32. For transformer-based architectures, soft calibration (INT8 Weak) is preferable over aggressive hard calibration, as shown in Table 17.

Sparsification without dedicated hardware support yields negligible computational savings. On the NVIDIA platform, where 2:4 structured sparsity is hardware-accelerated, meaningful throughput improvements are achievable for convolutional models.

Pruning delivered modest improvements unless the model was substantially oversized relative to the task. Knowledge distillation consistently outperforms magnitude-based pruning as a model compression strategy when a smaller target architecture is acceptable.

Network customisation: adapting the operator mix to the specific constraints and strengths of the tar-

get hardware remains the highest-leverage optimization strategy for dedicated NPUs. Replacing operator types outside the ASIC’s supported set, or restructuring layers to maximize pipeline utilization on the Hailo-8, can mean the difference between a **FAIL** and a fully functional high-throughput deployment.

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